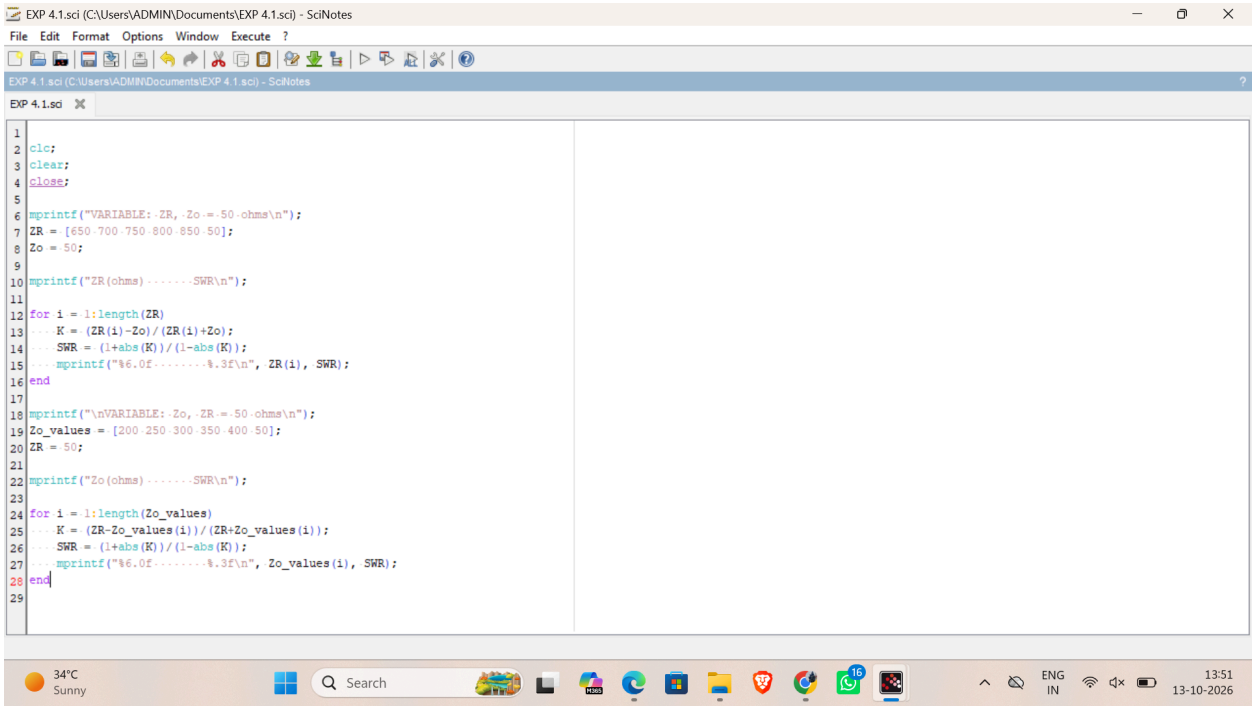
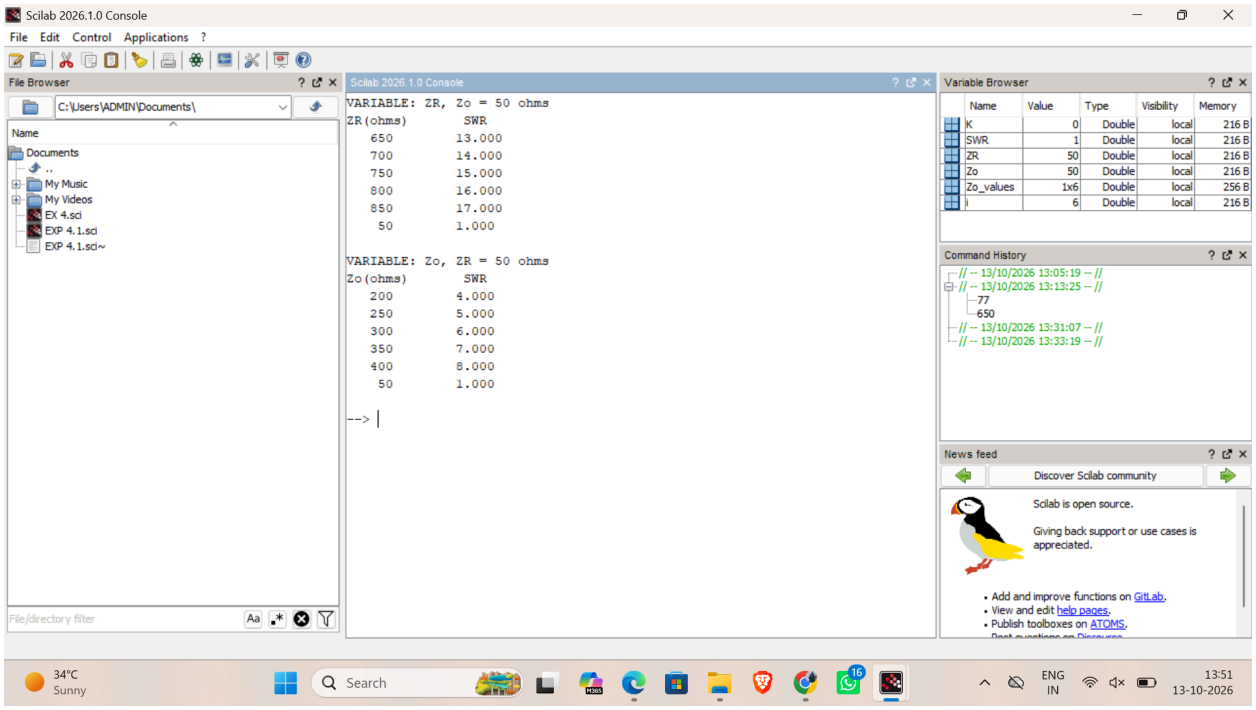
EXNO: 4 Analysis of SWR (Standing Wave Ratio)Transmission Lines by varying DATE: a)load im edance b) characteristic im edance

Test Case 1: Analyze the SWR of a transmission line for a fixed characteristic impedance (Zo) of 50 ohms, as the load impedance (ZL) is varied across 650 ohms to 850 ohms. Verify how the SWR changes with each value ofZL. When ZL=ZO, the SWR should equal 1 (indicating no reflection)

CODE:

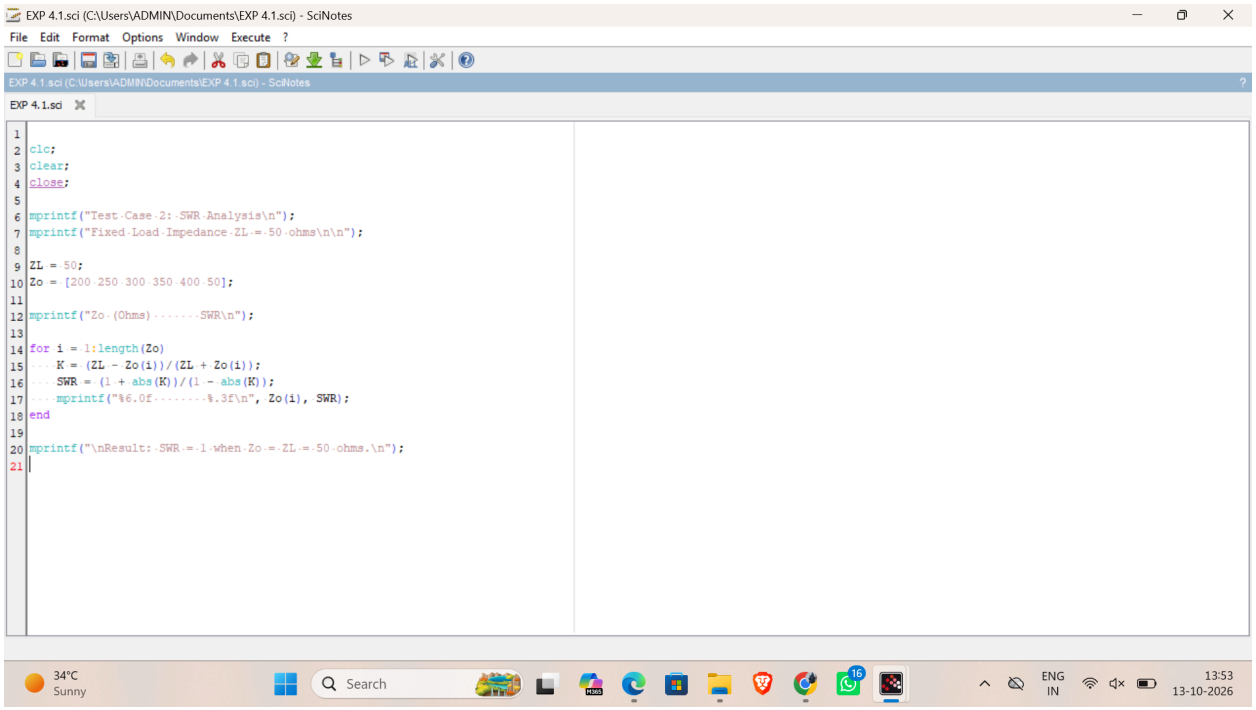


OUTPUT:

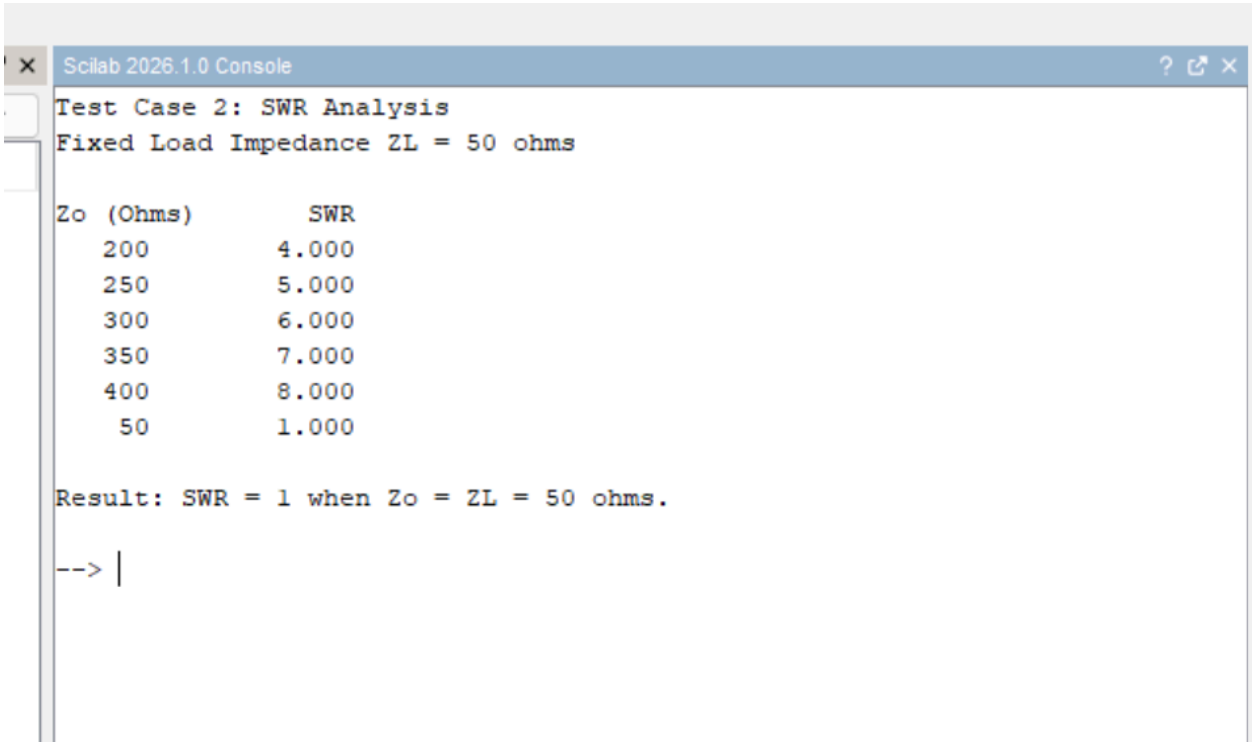


Test Case 2: Analyze the SWR for a fixed load impedance (ZL) of 50 ohms while varying the Scanned with OKEN Scanner characteristic impedance (Zo) of the transmission line to 250 ohms to 400 ohms. Verify how the SWR changes with each value Zo including Zo=ZL

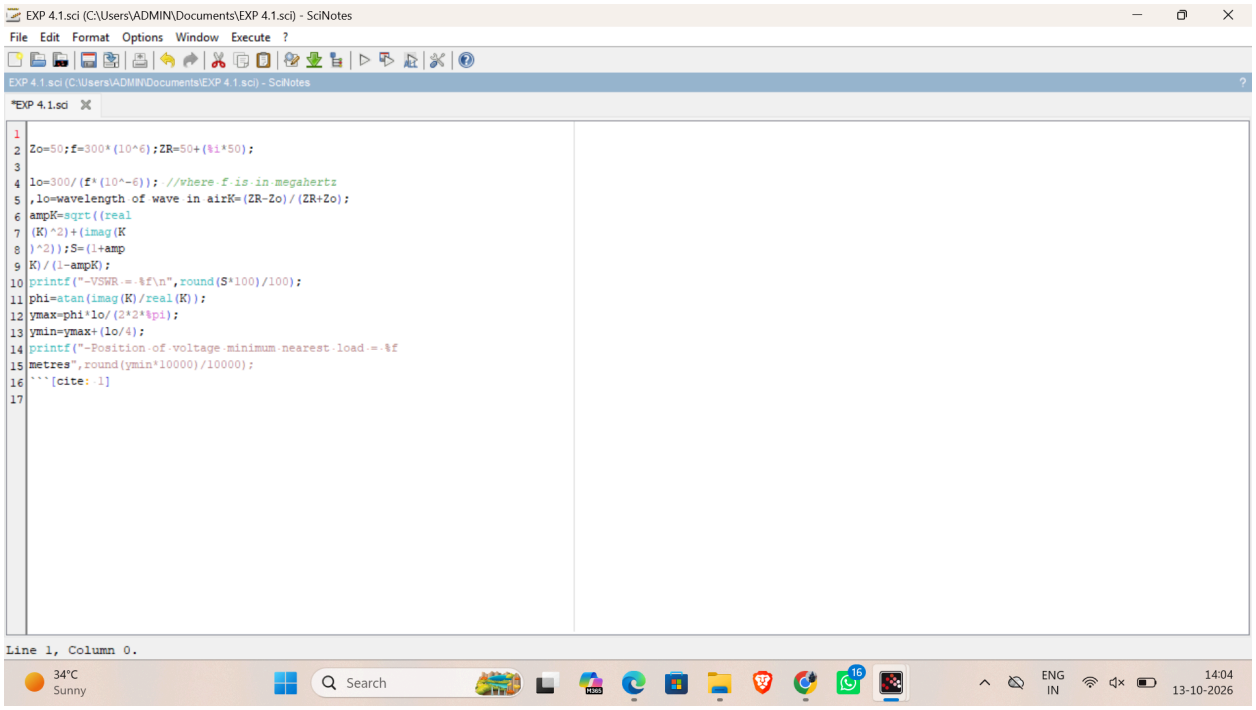
CODE:



OUTPUT:



EXNO: 5 DATE: Exploration of the relationship between position of Voltage minima nearest to he load by varying magnitude of reflection coefficient K
Test Case 1: For a fixed characteristic impedance (Zo) of 50 ohms and varying reflection coefficient magnitudes (KAE determine the position of the voltage minima (dm,n) relative to the load. The position of voltage minima should shift based on K and the corresponding phase of r. Compute (Imin using where r=KeJO
CODE:



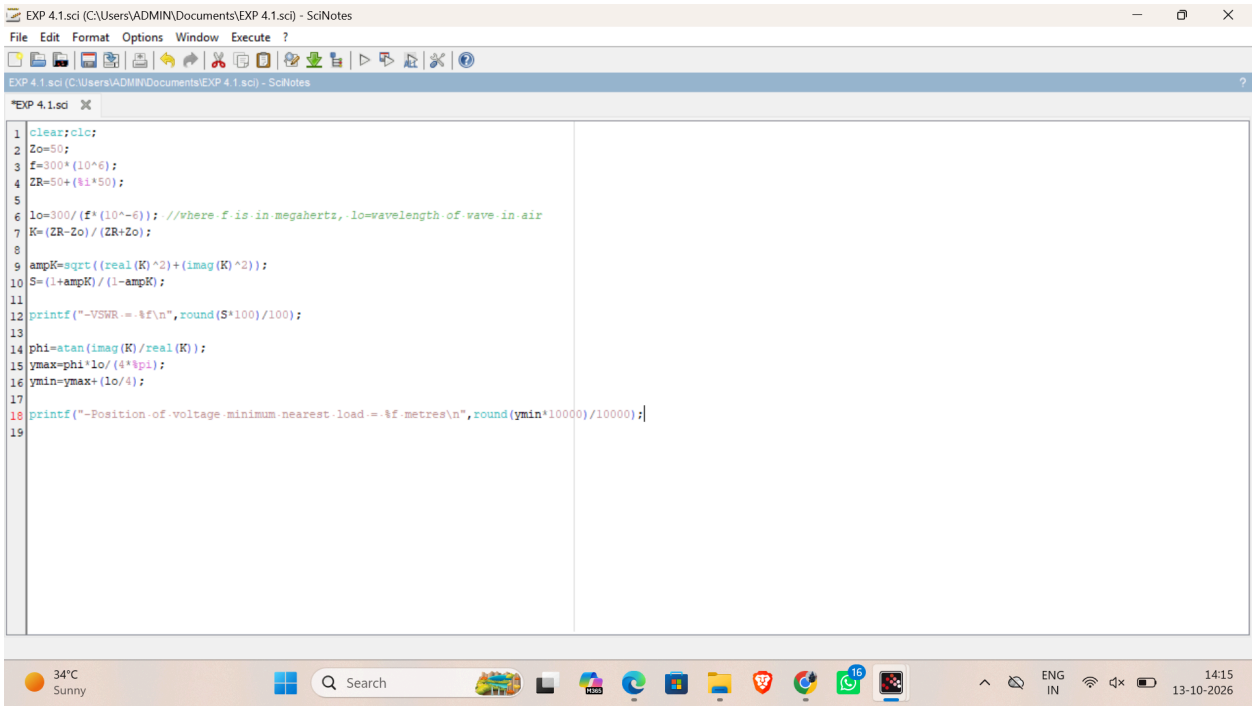
OUTPUT:

```
--> exec('C:\Users\ADMIN\Documents\EXP 4.1.sci', -1)
Variable: ZR Real Part
ZROhm    SWR      Position of Vmin (Simulation & Theoretical)
50       2.62     0.3381
60       2.46     0.3253
70       2.41     0.3133
80       2.44     0.3028

Variable: Zo
ZoOhm    SWR      Position of V min (Simulation & Theoretical)
50       2.62     0.3381
100      2.62     0.1619
150      3.37     0.1936
200      4.27     0.2087
```

Test Case 2: For a fixed reflection coefficient magnitude (K=O.5) and varying load impedances (ZL={25 Q,50 Q,IOO Q,150 Q}), calculate the position of voltage minima nearest to the load. Validate that the position of Vmin changes with the phase angle of r, which is influenced by the load impedance

CODE:



OUTPUT:

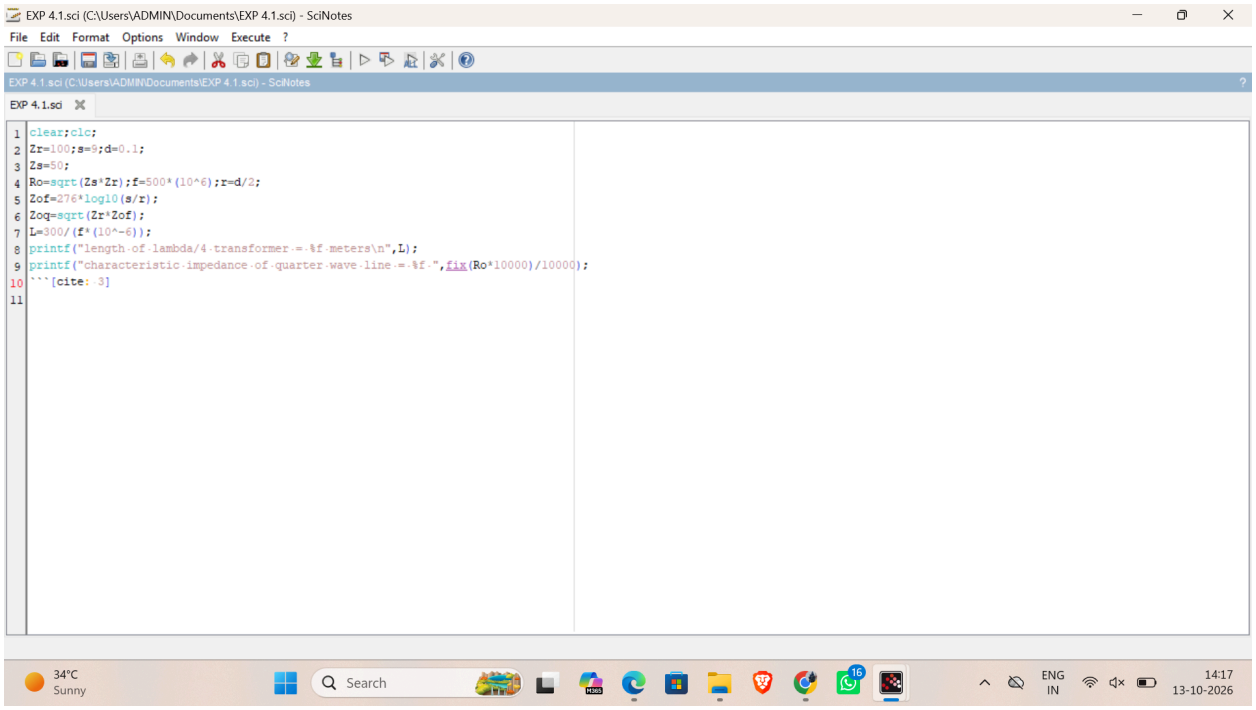
```
--> exec('C:\Users\ADMIN\Documents\EXP 4.1.sci', -1)
Test Case 2: Varying Load Impedance
-----------------------------------------------------------
ZL(Ohm)  |K|      Phase(rad)       Position of Vmin (m)
-----------------------------------------------------------
25       0.33     3.14             0.5000
50       0.00     0.00             0.2500
100      0.33     0.00             0.2500
150      0.50     0.00             0.2500
-----------------------------------------------------------

-->
```

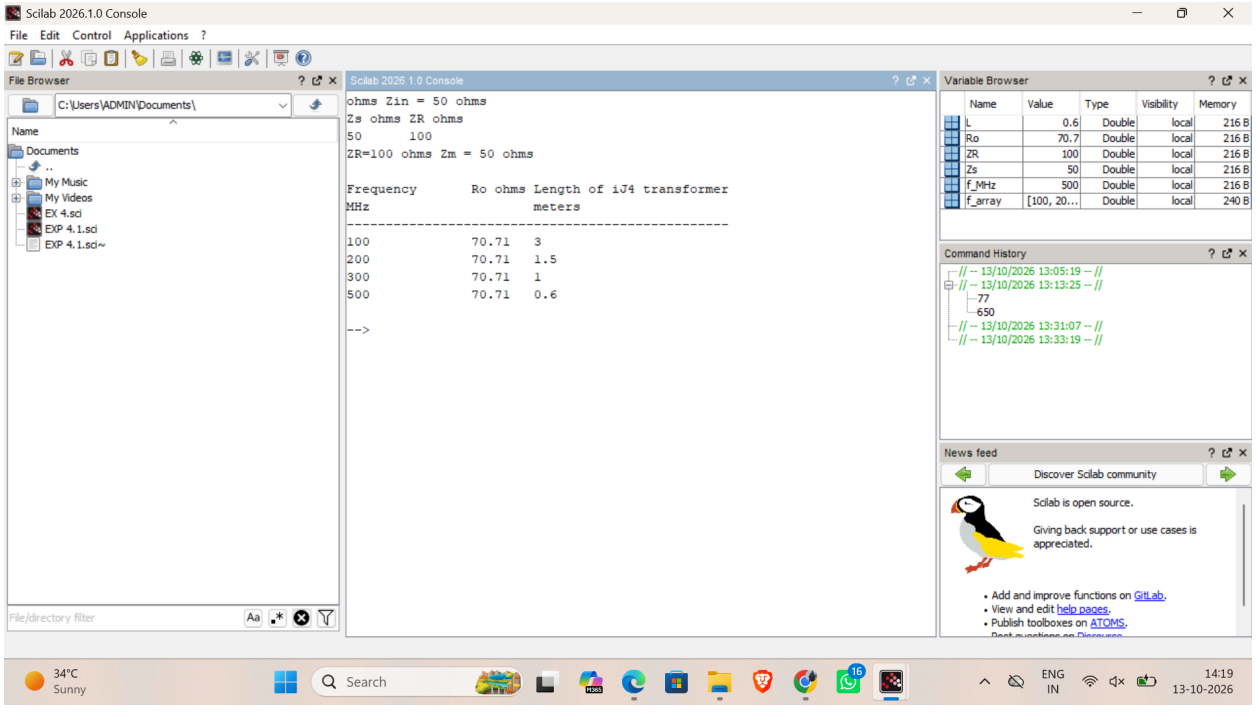
EXNO: 6 Designing a Quarter wave transformer by varying following parameters a) frequency b) input impedance c) characteristic impedance DATE:

Designing a Quarter wave transformer by varying following parameters d) frequency e) input impedance f) characteristic impedance Experiment Number:7 Test Case 1: Design a quarter-wave transformer for an input impedance (Zin) of 50 ohms and a load impedance (ZI) of 100 ohms. Analyze the transformer's performance as the operating frequency is varied across {100 MHz, 200 MHz, 300 MHz, 500 MHz}

CODE:



OUTPUT:



Test Case 2: For a fixed operating frequency of 2 GHz and a load impedance (ZL) of 100 ohms, design a quarter-wave transformer and evaluate its performance as the input impedance (Zin) varies through {25 ohms, 50 ohms,

75 ohms, 100 ohms}. Validate that the transformer's characteristic impedance (Ztransformer = Ro= changes with Zin

CODE:

EXP 4.1.sci (C:\Users\ADMIN\Documents\EXP 4.1.sci) - SciNotes

EXP 4.1.sci

```
clear;clc;
Zr=100;s=9;d=0.1;
Zs=50;
Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;
Zof=276*log10(s/r);
Zoq=sqrt(Zr*Zof);
L=300/(f*(10^-6));
printf("length of lambda/4 transformer = %f lambda\n",L);
printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);

```

OUTPUT:

Scilab 2026.1.0 Console

```
ZR=100 ohms Variable :Zin
Zin      Ro
-----------------
25       50.00
50       70.71
75       86.60
100      100.00

-->
```